

Expected value

Expected value is the probability-weighted mean of a random variable.

x	$P(X = x)$
0	0.1
1	0.3
2	0.4
3	0.2

What is the expected number of downloads in this synthetic model?

The weighted mean formula

Each possible value is weighted by its probability.

$$\mu = E(X) = \sum xP(X = x)$$

Why should a value with probability 0.4 receive more weight than one with probability 0.1?

Not an unweighted average

Synthetic model: $X = 0, 1, 2, 3$ with probabilities 0.1, 0.3, 0.4, 0.2.

Why is $(0 + 1 + 2 + 3)/4$ not the expected value?

A mean between integers

Synthetic model: $X = 0$ with probability 0.7 and $X = 1$ with probability 0.3.

Can the expected value be 0.3 even though X never equals 0.3?

Interpret the mean

A synthetic download-count model has expected value 1.7 files per session.

What does 1.7 describe over many sessions?

A short sequence

An independent repeated experiment has expected value 1.7 downloads per session.

Must ten sessions have exactly 17 downloads?

Law of large numbers for averages

With independent repetitions and a finite mean, the observed average tends toward the expected value as the number of repetitions grows.

Why can a game's short-run average differ greatly from its expected gain?

Expected total

A synthetic model has expected value 1.7 downloads per session.

What total would be expected over 100 sessions?

Variance of a distribution

Variance is the probability-weighted mean squared distance from μ .

$$\sigma^2 = \sum (x - \mu)^2 P(X = x)$$

What units would variance have if X measures dollars?

A deviation from expectation

Synthetic model: $X = 0$ or 2 , each with probability 0.5 ; $\mu = 1$.

What squared deviation belongs to each possible value?

Calculate a model variance

Synthetic model: $X = 0$ or 2 , each with probability 0.5 ; $\mu = 1$.

What is its variance?

Calculate model standard deviation

Synthetic model: $X = 0$ or 2 , each with probability 0.5 ; $\mu = 1$.

$$\sigma = \sqrt{\sum (x - \mu)^2 P(X = x)}$$

What is its standard deviation?

Unequal probability weights

Synthetic model: $P(X = 0) = 0.2$, $P(X = 1) = 0.5$, $P(X = 2) = 0.3$; $\mu = 1.1$.

What standard deviation results when each squared deviation receives its probability weight?

Same mean, different spread

Synthetic model A: X is always 1. Model B: $X = 0$ or 2 , each with probability 0.5 .

Which model has larger standard deviation?

Model versus sample

One statistic summarizes observed session counts; another summarizes their proposed probability distribution.

Which uses s , and which uses σ ?

Net gain

A fictional game costs \$2. A win pays a \$10 prize; a loss pays nothing.

What are the possible net gains?

Expected net gain

A fictional game costs \$2; a \$10 prize occurs with probability 0.1 and otherwise pays nothing.

What is the expected net gain per play?

Fair entry fee

A fictional game pays a \$10 prize with probability 0.1 and otherwise pays nothing.

What entry fee gives expected net gain zero?

A negative expected gain

A fictional game has expected net gain
−\$0.80 per play.

**Does this prevent a player from winning
on one play?**

Changing the variable

A fictional game's gross prize is X dollars; its entry fee is \$2.

How does the expected net gain compare with $E(X)$?

Shift and spread

Every possible prize increases by \$5.

How does this shift change the standard deviation of the prize?

A weighted average beyond games

Synthetic server selection: A is chosen with probability 0.4 and takes 2 seconds; B with probability 0.6 takes 5 seconds.

What is the expected response time?